Ezra **Klukas**

UBC Engineering Physics Student · Honours Mathematics Minor

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Education

BASc in Engineering Physics, Honours Mathematics Minor — *Vancouver, BC*

UBC — *Sep. 2023 - Apr. 2028*

- Electrical, software, computer, and mechanical introductory engineering courses, large scale team technical project courses, undergirded by rigorous foundation in pure and applied mathematics and physics
- Textbook driven emphasis on pure math courses, such as group theory and graph theory, and many more algebra and analysis courses to come.
- 89.7% average, awarded TREK Excellence Scholarship for top 5% of domestic students (2024, 2025), and Donald J. Evans Scholarship in Engineering, and EXPO 86 Scholarship awarded to academically outstanding engineering students

Experience

UBC Sensorimotor Physiology Lab - Dr. Jean-Sébastien Blouin & Jesse Charlton — *Vancouver, BC*

Research Engineer Co-op — *May. 2026 - Aug. 2026*

- Developing the embedded core of a modular real-time human-device experiment platform, integrating high-density EMG acquisition, online motor-unit decomposition, and deterministic EtherCAT servo control on an NVIDIA Jetson Orin Nano.
- Implemented real-time EtherCAT control for a Teknic ClearPath CiA-402 servo in C/C++ using SOEM and IgH, including PDO mapping, CSP mode setup, operational-state transitions, distributed-clock timing, and cyclic diagnostics.
- Configured PREEMPT_RT Linux for hard real-time control, using CPU isolation, IRQ affinity, scheduler tuning, and load testing to profile latency, jitter, deadline margin, and kHz-rate control-loop feasibility.
- Designed and evaluated FPGA, MCU, and Linux-based acquisition architectures for scalable Intan RHD2000 EMG streaming over custom SPI/DDR-SPI, with attention to timing closure, timestamping, buffering, synchronization, and host transfer over Ethernet/TCP/UDP.
- Built MATLAB/C++ tooling for the offline-to-online EMG pipeline, including decomposition validation, motor-unit filter export, YAML configuration, binary data handling, and deployment planning for real-time signal processing.
- Developed practical fluency with embedded Linux, C/C++, Git, vi/vim, gdb-style debugging workflows, oscilloscopes, logic-analyzer reasoning, and protocols including SPI, EtherCAT, Ethernet, TCP/UDP, I2C, and UART.

Siemens Mobility Austria GmbH — *Vienna & Graz, Austria*

Project Management Intern — *Jul. - Aug. 2023, Jul. - Aug. 2024*

- Supported quality engineering and project-management workflows for OBB night-train projects, inspecting reported issues, assessing severity, and coordinating next-step resolution paths.
- Built Python and Office Script automation tools for weekly progress analysis, quality-issue tracking, form generation, and ISO documentation workflows across Austrian and Czech teams.
- Conducted 30+ stakeholder interviews to catalogue internal software tools, superusers, license metadata, and transition requirements, improving continuity during software/process changes.

Technical Projects

Self-driving Autonomous Launching Robot — *UBC, ENPH 253*

C++, ESP32, KiCAD Schematic & PCB design, Oscilloscope and DMM Debugging, Circuit analysis & design, EMF noise troubleshooting, CAD prototyping, systems integration — *May - August 2025*

- Collaborated in a team with three other outstanding students to design, prototype, build, and test an autonomous self-driving robot from scratch, for competition.
- Designed, printed, and soldered working H-Bridge motor driver PCBs.
- Designed a torsion spring-loaded throwing arm, with a working 3D printed and machined servo activated clutch to engage and disengage winding motor, and release the arm.
- Implemented and designed LiDaR sensing algorithms, and heavy involvement in systems integration and code development in C++, using an ESP32 microprocessor to control the robot.

Simulated Autonomous Driving Competition — *UBC, ENPH 353*

Python, TensorFlow, Keras, ROS, Gazebo, OpenCV, GymGazebo, Linux, Bash — *Sep. - Dec. 2025*

- Developed artisanal and CNN imitation learned autonomous self driving algorithm for car in ROS Gazebo simulated environment, for class competition.
- Developed and optimized light-weight plate detection, character finding, and CNN character reading algorithm with seamless passive driving integration.
- Implemented DAGGR with QtGUI interface to optimize CNN imitation learning corrective behaviour.